# Chained ECDLP transfers over prime fields

# Chained ECDLP transfers over prime fields — verified construction, measured boundary

**Status:** research construction + measurement. Same characteristic only. No cryptanalytic speedup; no cross-characteristic claim. Every number here is `[pcc:unverified]` until CI's pcc gate signs the claims.

Protocol frozen in `PROTOCOL.md` (`frozen.sha256`) before any measurement. Builds on PR #3877 (toy genus-2 transfer), promoted to native Rust.

## What this adds over PR #3877

PR #3877 proved `f_*f^* = [2]` algebraically and checked a toy `F_101` instance in Python. This effort:

1. **Native Rust, arbitrary divisors.** `src/cryptanalysis/chained_transfer/` implements `F_p`/`F_{p²}`, elliptic curves with Vélu isogenies, and the bielliptic quotient maps `φ^*` (pull-back) and `φ_*` (push-forward) on **arbitrary reduced Mumford divisors**, including non-split ones (roots over `F_{p²}`), built on the repository's `genus2_fp` Cantor arithmetic. This closes the "arbitrary nonsplit divisor" gap named in `problems/prime-cover-balanced-jacobian-and-induced-maps-missing.md`.
2. **Verified chains.** A transfer *hop* checks `h(P) ≠ 0`, `ord = r`, and `h([x]P) = [x]h(P)` before it is trusted; a *chain* `E → E' → J(C)` (isogeny then pull-back) is verified end to end to preserve the order-`r` discrete logarithm.
3. **Measured total cost.** `chained_transfer_bench` solves the transferred instance four ways and compares counted group operations against a matched Pollard-rho reference on the source curve.

## Results

See `RESULTS.md` (generated) and `results.json`. Summary: *(filled by the run)*.

## Reproduce

```
cargo test  --release --lib chained_transfer
cargo run   --release --bin chained_transfer_bench -- run --json \
    research/chained_prime_transfer_20261111/results.json
```

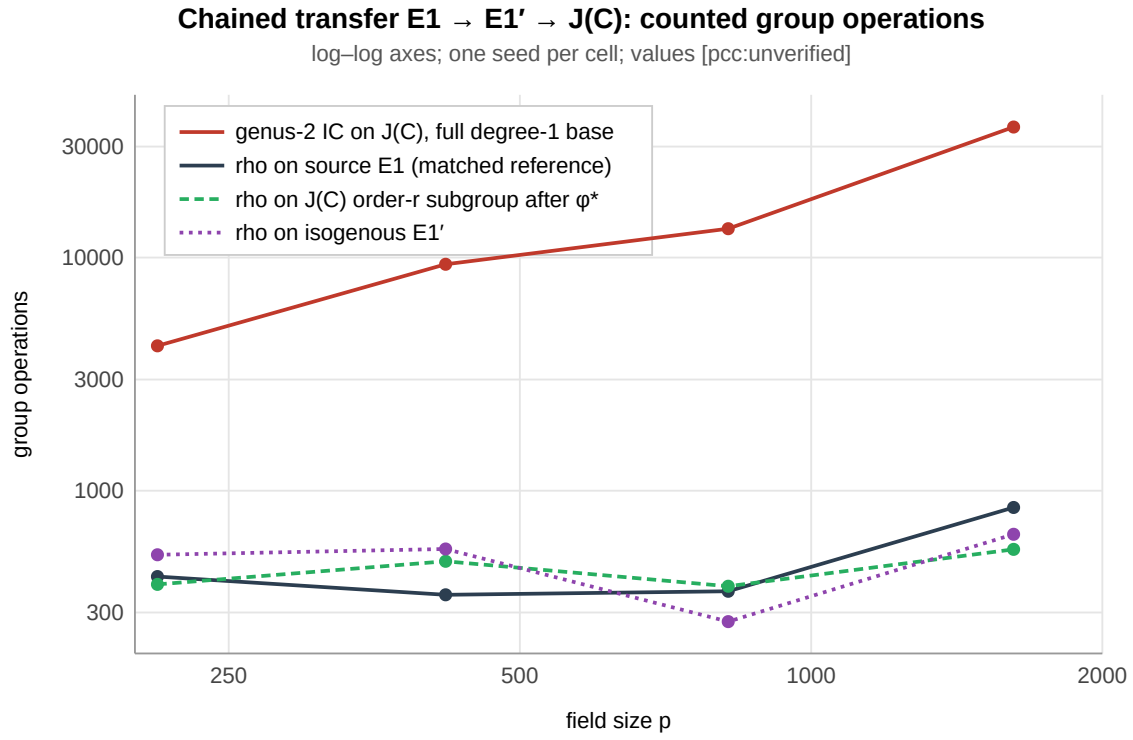
## Scope and obstruction

The genus-2 destination is not cheaper: see `problems/prime-field-chained-transfer-no-advantage.md`. The boundary is the `Ω(p)` relation count of genus-2 index calculus over `F_p` against rho's `√p` — a structural obstruction, not a tuning gap. # Results: chained prime-field transfer, measured boundary

Run: `chained_transfer_bench run` (bench record `benchmarks/records/20261111T0540Z-chained-prime-transfer.json`), deterministic. A second run under the record reproduced every count exactly. Local development check (exact operation counts, not timings); one seed per cell. All values are `[pcc:unverified]`.

| p | r | rho on E1 (reference) | rho on E1′ (isogeny, deg) | rho on J(C) after φ* | IC on J(C) | IC trials | IC / rho |
|---|---|---|---|---|---|---|---|
| 211 | 109 | 428 | 530 (2) | 396 | 4,184 | 244 | 9.8× |
| 419 | 113 | 357 | 562 (2) | 497 | 9,366 | 532 | 26.2× |
| 821 | 71 | 370 | 274 (3) | 389 | 13,313 | 869 | 36.0× |
| 1619 | 401 | 847 | 649 (2) | 560 | 36,363 | 1,588 | 42.9× |

Units: counted group operations (elliptic additions and doublings for rho on `E1`/`E1′`; Cantor additions and doublings for `J(C)`). The IC column counts relation collection only; linear algebra over `Z/r` is extra and is not included, so the IC figure is a lower bound on its total cost.




cost vs p

# What the run establishes

1. **The chain is a valid DLP transfer at every size.** Each hop passed the in-code admissibility checks (`h(P) ≠ 0`, `ord h(P) = r`, `h([x]P) = [x]h(P)` on sampled `x`), the composed chain was verified end to end, and the index calculus recovered the known log in every cell.
2. **F1 (rho advantage) — not met.** Rho on the isogenous curve and on the order-`r` subgroup of `J(C)` lands within the run-to-run scatter of rho on the source (single seed per cell, so no tighter statement is made). This is expected: isogenous curves have the same order, and the `J(C)` subgroup is still cyclic of order `r`. Per group operation, a Cantor step costs more field work than an elliptic addition, so the field-operation comparison only favours the source.
3. **F2 (index calculus beats rho) — not met.** Genus-2 IC on `J(C)` costs 9.8× to 42.9× the matched rho, and the ratio grows with `p`.
4. **F3 (IC exponent).** A least-squares fit of ln(IC group operations) against ln p over the four sizes gives slope **1.01** (derived; four points, no interval). This is the `Ω(p)` relation count of genus-2 index calculus (about `p/2` degree-1 places, smoothness about 1/2, so about `p` trials). Rho scales as `√r ≤ √p`. The gap widens as `p` grows: the boundary is structural, not a tuning constant.

# What it does not establish

- No timing claim; no isolation tier; no validation-gate admission.
- One bielliptic curve per size, one seed per cell; no spread for the rho columns.
- Nothing about other cover families, genera above 2, or cross-characteristic maps (none is constructed).